

Q1 -3
Q3 -2 calculation errors
Q5 -3
Q2 Good
Q4 -0

Homework Assignment 3

Show all your computations and justify your answers.

1. (20 points) Find the numbers at which f is discontinuous. At which of these numbers is f continuous from the right, from the left, or neither? Sketch the graph of f .

$$(a) f(x) = \begin{cases} e^x & \text{if } x \leq 0, \\ -(x-1)^2 + 2 & \text{if } 0 < x < 2, \\ \ln(x-1) + 2 & \text{if } x \geq 2. \end{cases} \quad (b) f(x) = \begin{cases} \sin^2(2x) & \text{if } x \leq -\frac{\pi}{2}, \\ \frac{4x^3 - 4x^2 - \pi^2 x + \pi^2}{x-1} & \text{if } -\frac{\pi}{2} < x < \frac{\pi}{2}, \\ \tan(x - \frac{\pi}{2}) & \text{if } x \geq \frac{\pi}{2}. \end{cases}$$

(a) $\lim_{x \rightarrow 0^-} f(x) = e^0 = 1$, $\lim_{x \rightarrow 0^+} f(x) = -(0-1)^2 + 2 = 1$

$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0^+} f(x) \Rightarrow$ continuous

$\lim_{x \rightarrow 2^-} f(x) = -(2-1)^2 + 2 = 1$, $\lim_{x \rightarrow 2^+} f(x) = \ln(2-1) + 2 = 2$

$\lim_{x \rightarrow 2^-} f(x) \neq \lim_{x \rightarrow 2^+} f(x) \Rightarrow$ discontinuous

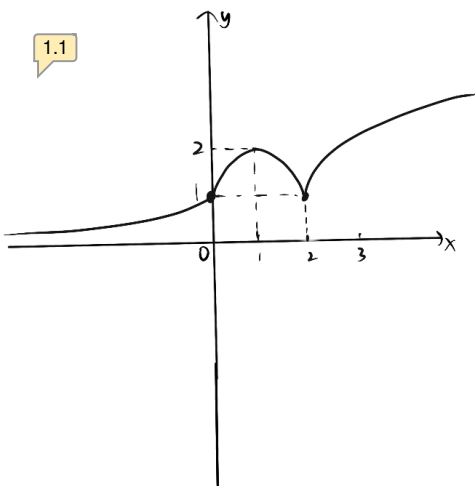
In conclusion, f is discontinuous when $x=2$.

f is continuous from the left of $x=2$

The graph:

X

1.1



(b) $\frac{4x^3 - 4x^2 - \pi^2 x + \pi^2}{x-1} = \frac{4x^2(x-1) - \pi^2(x-1)}{x-1} = 4x^2 - \pi^2 (x \neq 1)$

$\lim_{x \rightarrow -\frac{\pi}{2}^-} f(x) = \sin^2(2 \cdot (-\frac{\pi}{2})) = \sin^2(-\pi) = 0$

$\lim_{x \rightarrow -\frac{\pi}{2}^+} f(x) = 4 \cdot (-\frac{\pi}{2})^2 - \pi^2 = 0$

$\lim_{x \rightarrow -\frac{\pi}{2}} f(x) = \lim_{x \rightarrow -\frac{\pi}{2}^-} f(x) = \lim_{x \rightarrow -\frac{\pi}{2}^+} f(x) \Rightarrow \lim_{x \rightarrow -\frac{\pi}{2}} f(x)$ exist, continuous

$\lim_{x \rightarrow \frac{\pi}{2}^-} f(x) = 4 \cdot (\frac{\pi}{2})^2 - \pi^2 = 0$

$\lim_{x \rightarrow \frac{\pi}{2}^+} f(x) = \tan(\frac{\pi}{2} - \frac{\pi}{2}) = 0$

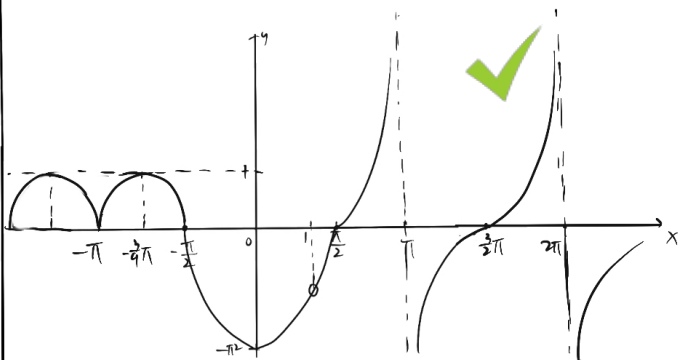
$\lim_{x \rightarrow \frac{\pi}{2}} f(x) = \lim_{x \rightarrow \frac{\pi}{2}^-} f(x) = \lim_{x \rightarrow \frac{\pi}{2}^+} f(x) \Rightarrow \lim_{x \rightarrow \frac{\pi}{2}} f(x)$ exist, continuous

③ Since $f(1)$ is not exist, $f(x)$ is discontinuous when $x=1$

④ $\tan(x - \frac{\pi}{2})$ domain is $x \neq \pi + k\pi (k \in \mathbb{Z})$

In conclusion, f is discontinuous when $x=1$ and $\pi + k\pi (k \in \mathbb{Z})$

f is continuous from the left of $x=1$



2. (20 points) Find the values of a and b that make f continuous everywhere.

$$f(x) = \begin{cases} \frac{x^3 - x^2 - x + 1}{x-1} & \text{if } x < 1, \\ ax^2 + bx + 1 & \text{if } 1 \leq x < \pi, \\ 2 \sin\left(x - \frac{5\pi}{2}\right) & \text{if } x \geq \pi. \end{cases}$$

To make f continuous everywhere, we should make

$\lim_{x \rightarrow 1^-} f(x)$ and $\lim_{x \rightarrow \pi^-} f(x)$ exist.

Since $\lim_{x \rightarrow a} f(x)$ exist $\iff \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x)$

$$\textcircled{1} \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} \frac{x^3 - x^2 - x + 1}{x-1} = \lim_{x \rightarrow 1^-} \frac{(x+1)(x-1)^2}{x-1} = \lim_{x \rightarrow 1^-} (x+1)(x-1) = 2 \cdot 0 = 0$$

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} ax^2 + bx + 1 = a + b + 1$$

Then we can get $a + b + 1 = 0$ (*)

$$\textcircled{2} \lim_{x \rightarrow \pi^-} f(x) = \lim_{x \rightarrow \pi^-} ax^2 + bx + 1 = a\pi^2 + b\pi + 1$$

$$\lim_{x \rightarrow \pi^+} f(x) = \lim_{x \rightarrow \pi^+} 2 \sin\left(x - \frac{5\pi}{2}\right) = 2 \sin\left(\pi - \frac{5\pi}{2}\right) = 2$$

Then we can get $a\pi^2 + b\pi + 1 = 2$ (x)

According to (x)

$$\begin{cases} a + b + 1 = 0 \\ a\pi^2 + b\pi + 1 = 2 \end{cases} \Rightarrow \begin{cases} a + b = -1 & \textcircled{1} \\ \pi(a\pi + b) = 1 & \textcircled{2} \end{cases}$$

we can get $a = -b - 1$ ③

put ③ in ②, $\pi[(-b-1)\pi + b] = 1$

$$-b\pi + b - \pi = \frac{1}{\pi}$$

$$b(1 - \pi) = \frac{1 + \pi^2}{\pi}$$

$$b = \frac{\pi^2 + 1}{\pi(1 - \pi)}$$

Then we can put b in ①, get $a = -1 - \frac{\pi^2 + 1}{\pi(1 - \pi)}$

In conclusion, $a = -1 - \frac{\pi^2 + 1}{\pi(1 - \pi)}$, $b = \frac{\pi^2 + 1}{\pi(1 - \pi)}$

3. (24 points) Find the vertical and horizontal asymptotes of the following curves.

(a) $f(x) = \frac{x-1}{x^2+x-1}$.

(a) $f(x) = \frac{x-1}{x^2+x-1}$

① First, we can find vertical asymptotes.

Assume $x^2+x-1=0$ then we get $x = \frac{1 \pm \sqrt{5}}{2}$

Since the domain of $f(x)$ isn't include $x = \frac{1 \pm \sqrt{5}}{2}$

The vertical asymptotes are $x = \frac{1 \pm \sqrt{5}}{2}$

② Then we find

horizontal asymptotes

When $x \rightarrow \pm\infty$, $f(x) = \frac{x-1}{x^2+x-1} \approx \frac{x}{x^2} = \frac{1}{x}$

So $\lim_{x \rightarrow \pm\infty} f(x) = \lim_{x \rightarrow \pm\infty} \frac{1}{x} = 0$

The horizontal asymptote is $y=0$

In conclusion

the vertical asymptotes are $x = \frac{1 \pm \sqrt{5}}{2}$

the horizontal asymptote is $y=0$

(b) $g(x) = \frac{\cos^2 x}{x^2+1} + 1$.

① Vertical asymptote

Since $x^2+1 > 1 \neq 0$, the domain is $\mathbb{R}$, that means there is no vertical asymptote

② horizontal asymptote

When $x \rightarrow \pm\infty$, since $\cos^2 x \in [0, 1]$, $x^2+1 \rightarrow \infty$, $\frac{\cos^2 x}{x^2+1} \rightarrow 0$

$\lim_{x \rightarrow \pm\infty} g(x) = \lim_{x \rightarrow \pm\infty} \frac{\cos^2 x}{x^2+1} + 1 = 0 + 1 = 1$

The horizontal asymptote is $y=1$

In conclusion, $g(x)$ doesn't have vertical asymptote, the horizontal asymptote is $y=1$



(c) ① vertical asymptote

Since $1+4e^{2x} > 1$, the domain is $\mathbb{R}$, that means there is no vertical asymptote

② horizontal asymptote

When $x \rightarrow \infty$, $e^{2x} \rightarrow \infty$, $h(x) = \frac{1+e^{2x}}{1+4e^{2x}} \approx \frac{e^{2x}}{4e^{2x}} = \frac{1}{4}$

When $x \rightarrow -\infty$, $e^{2x} \rightarrow 0$, $h(x) = \frac{1+e^{2x}}{1+4e^{2x}} \approx \frac{1}{1+0} = 1$

So the horizontal asymptote is $y=\frac{1}{4}$ and $y=1$

In conclusion, $h(x)$ doesn't have vertical asymptote, the horizontal asymptote is $y=1$ and $y=\frac{1}{4}$

4. (16 points)

- (a) For $x > 0$, prove that the derivative of $f(x) = \sqrt{x}$ is $f'(x) = \frac{1}{2\sqrt{x}}$ by applying the definition of derivative as a limit.
- (b) Explain why $f'(0)$ does not exist.

(a) Prove

According to the definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

$$\begin{aligned} \text{To } f(x) = \sqrt{x}, \quad f'(x) &= \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \times \frac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}} \\ &= \lim_{h \rightarrow 0} \frac{(x+h) - x}{h(\sqrt{x+h} + \sqrt{x})} \\ &= \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+h} + \sqrt{x}} \\ &= \frac{1}{\sqrt{x} + \sqrt{x}} \\ &= \frac{1}{2\sqrt{x}} \end{aligned}$$

(b) Prove

$$\begin{aligned} f'(0) &= \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sqrt{h} - \sqrt{0}}{h} \end{aligned}$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{h}}{h} = \lim_{h \rightarrow 0} \frac{1}{\sqrt{h}}$$



Since $\lim_{h \rightarrow 0} \frac{1}{\sqrt{h}} = \infty$, this limit doesn't exist

That means $f'(0)$ doesn't exist.

In conclusion, the reason is when $h \rightarrow 0$

$\lim_{h \rightarrow 0} \frac{\sqrt{h}}{h}$ doesn't exist.

5. (20 points) Find a , b , c and d in $\mathbb{R}$ such that the following function is differentiable.

$$f(x) = \begin{cases} \sin x & \text{if } x \leq 0, \\ ax^3 + bx^2 + cx + d & \text{if } 0 < x < 1, \\ -x^2 + 2x & \text{if } x \geq 1. \end{cases}$$

Is f' differentiable?

$f(x)$ is differentiable when $f(x)$ is continuous in $x=0$ and $x=1$ and the derivatives from the left and right must equal at these points.

$$\lim_{x \rightarrow a} f(x) \text{ exists} \iff \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x)$$

(1) $x=0$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \sin x = \sin 0 = 0$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} ax^3 + bx^2 + cx + d = d$$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) \rightarrow d = 0 \quad \textcircled{1}$$

$$\lim_{x \rightarrow 0^-} f'(x) = \lim_{x \rightarrow 0^-} \cos x = 1$$

$$\lim_{x \rightarrow 0^+} f'(x) = \lim_{x \rightarrow 0^+} 3ax^2 + 2bx + c = c$$

$$\lim_{x \rightarrow 0^-} f'(x) = \lim_{x \rightarrow 0^+} f'(x) \rightarrow c = 1 \quad \textcircled{2}$$

(2) $x=1$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} ax^3 + bx^2 + cx + d = a + b + c + d$$

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} -x^2 + 2x = 1$$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x) \rightarrow a + b + c + d = 1 \quad \textcircled{3}$$

$$\lim_{x \rightarrow 1^-} f'(x) = \lim_{x \rightarrow 1^-} 3ax^2 + 2bx + c = 3a + 2b + c$$

$$\lim_{x \rightarrow 1^+} f'(x) = \lim_{x \rightarrow 1^+} -2x + 2 = 0$$

$$\lim_{x \rightarrow 1^-} f'(x) = \lim_{x \rightarrow 1^+} f'(x) \rightarrow 3a + 2b + c = 0 \quad \textcircled{4}$$

According to ①②③④

$$\begin{cases} d=0 \\ c=1 \\ a+b+c+d=1 \\ 3a+2b+c=0 \end{cases} \Rightarrow \begin{cases} a=-1 \\ b=1 \\ c=1 \\ d=0 \end{cases}$$

$$f(x) = \begin{cases} \sin x & x \leq 0 \\ -x^3 + x^2 + x & 0 < x < 1 \\ -x^2 + 2x & x \geq 1 \end{cases}$$

Then we should prove $f'(x)$ is differentiable

In past steps we have already shown that

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x), \quad \lim_{x \rightarrow 0^-} f'(x) = \lim_{x \rightarrow 0^+} f'(x)$$

$$\text{Also } f'(x) = \begin{cases} \cos x & x \leq 0 \\ -3x^2 + 2x + 1 & 0 < x < 1 \\ -2x + 2 & x \geq 1 \end{cases}$$

is differentiable in all intervals.

In conclusion $f'(x)$ is differentiable

Index of comments

- 1.1 incorrect, f is continuous from the right at $x=2$
 -2
 incorrect graph -1